

DLD – Quiz

Time: 11:30am – 2:15pm – No break

Quiz – 1

1. Conver the following:
 - a. -28_{10} to binary
 - b. 12.34_{10} to binary
 - c. $A34.2A_{16}$ to binary
 - d. 101011.11_2 to Hexadecimal

[1]

2. Draw a truth table for the following function:

$$G = X'Y + (X + Z')(Y + Z)$$

[1]

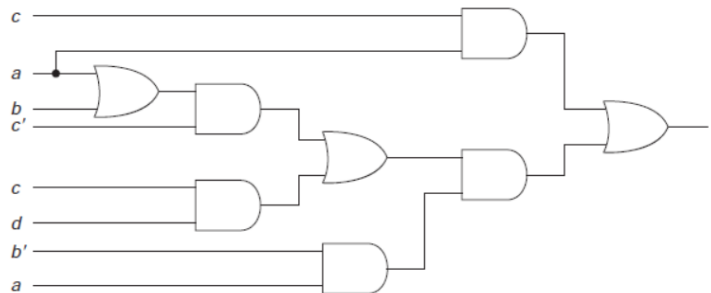
3. Reduce the following expression to a minimum SOP form using Boolean Algebra reduction.

$$x'y'z' + x'y'z + x'yx + xy'z + xyz$$

Quiz – 2

[1]

4. For the following circuit, (i) find the algebraic expression (ii) write it in SOP form



[2]

5. Find the complement of the following expression.

$$g = (w + x' + y)(w' + x + z)(w + x + y + z)$$

[1]

6. For the following truth table:

Input			Output
a	b	c	f
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	1

- a. Write the minterms in numerical form.

- b. Write the SOP expression.
- c. Write the maxterms in numerical form.
- d. Write the POS expression.

[2]

7. For the following expression, draw the circuit using only NAND gates:

$$f = ab'd' + bde' + bc'd + a'ce$$

[1]

8. For the following expression, draw the circuit using only NOR gates:

$$f = (a + b')(a' + c + d)(b + d')$$

[1]

Quiz 3

9. Using K-Maps, find the minimum SOP expression the following functions:

- a. $f(w, x, y, z) = \sum m(2, 3, 5, 7, 10, 13, 14, 15)$
- b. $f(a, b, c, d) = \sum m(3, 4, 9, 13, 14, 15) + \sum d(2, 5, 10, 12)$

[2]

10. Consider a combinational logic system that determines if a 4-bit binary quantity A,B,C,D that is divisible by decimal numbers two, three, seven or nine with no remainder

- a. Draw the truth tables for Dby2(A,B,C,D), Dby3(A,B,C,D), Dby7(A,B,C,D), Dby9(A,B,C,D)

	A	B	C	D	Dby2	Dby3	Dby7	Dby9
0	0	0	0	0				
1	0	0	0	1				
2	0	0	1	0				
3	0	0	1	1				
4	0	1	0	0				
5	0	1	0	1				
6	0	1	1	0				
7	0	1	1	1				
8	1	0	0	0				
9	1	0	0	1				
10	1	0	1	0				
11	1	0	1	1				
12	1	1	0	0				
13	1	1	0	1				
14	1	1	1	0				
15	1	1	1	1				

- b. Write the SOP expression.
- c. Minimize using K-Maps to find minimized Sum of Products forms.
- d. Draw the logic circuit diagram

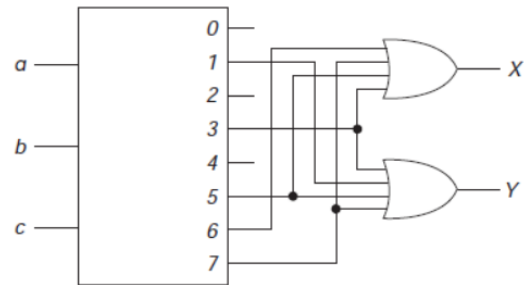
[3]

Quiz 4

11. Design a 4-bit adder subtractor which can do both addition and subtraction.

[2]

12. Consider the following circuit with an active high output (that is the circuit performs its function when the output is 1) decoder. Draw the truth table for the outputs:



13. Design an eight-way multiplexer using 2-way multiplexers.

[1]

[2]